

DEPARTMENT OF COMPUTER SYSTEMS
ENGINEERING

Visual Breakdown HCI vs. Computer Graphics

Mini Assignment Deconstruction Project

Student Name: Mohammed Khayyam

Roll Number: 2k24/CSE/81

Subject: Computer Graphics

1. Objective and Introduction

The objective of this assignment is to demonstrate a thorough understanding of the differences and intersections between Human-Computer Interaction (HCI) and Computer Graphics (CG). This is achieved by visually annotating and analyzing an interactive 3D digital tool [cite: 1].

Software Selection: Google Earth

For this analysis, I have chosen **Google Earth**. Google Earth is a highly interactive 3D application that relies heavily on advanced computer graphics to render the globe using satellite imagery, aerial photography, and GIS data onto a 3D globe [cite: 1]. Simultaneously, it features a complex yet intuitive user interface designed through HCI principles to allow users to seamlessly navigate a massive 3D environment.

The breakdown provided in the subsequent pages will isolate the specific elements that belong to the realm of Computer Graphics (rendering, lighting, geometry) and separate them from the elements belonging to Human-Computer Interaction (buttons, controls, feedback systems) [cite: 1].

By evaluating Google Earth, we can clearly see how high-fidelity graphics rely on intuitive interface controls to make a robust digital tool accessible to the average user.

2. HCI Focus (Human-Computer Interaction)

Human-Computer Interaction (HCI) focuses on how users interact with the software. In a 3D application, the UI must provide clear affordances so the user understands how to manipulate the 3D space without feeling disoriented.

[Screenshot 1: Google Earth UI & Interaction Elements]

(Visualizing Search Bar, Navigation Compass, and Info Panels) [cite: 1]

Annotations & Usability Analysis

1. On-Screen Navigation Controls (Compass & 3D Toggle)

Interaction Element: Located at the bottom right of the screen, these include a compass icon, zoom buttons (+/-), and a 2D/3D toggle switch [cite: 1].

Usability Purpose: These tools provide direct, click-based alternatives to mouse-drag gestures. They offer clear visual affordances for spatial manipulation. If a user gets lost or disoriented while rotating the globe, clicking the compass instantly re-orientes the view to face North, functioning as an essential "reset" usability feature [cite: 1].

2. Floating Search Bar & Main Menu

Interaction Element: A minimalist search bar located at the top-left corner, paired with a hamburger menu.

Usability Purpose: This provides an alternative interaction paradigm. Instead of manually scrolling and zooming across the globe (which can be tedious), the search bar allows users to input text and rely on the software to automatically navigate to the location. This severely reduces cognitive load and physical effort for the user.

3. Dynamic Knowledge Cards (Contextual Feedback)

Interaction Element: When a specific landmark is clicked or searched, a 2D information panel slides in over the 3D view [cite: 1].

Usability Purpose: This serves as dynamic feedback text. It bridges the gap between spatial exploration and factual data gathering. It helps the user by providing immediate context (population, history, photos) without forcing them to leave the 3D environment or open a new browser tab [cite: 1].

3. CG Focus (Computer Graphics)

Computer Graphics (CG) deals with the programmatic rendering of visual elements. It is responsible for taking raw data and mathematical coordinates and converting them into the visual 3D representation that the user sees on the screen.

[Screenshot 2: Google Earth 3D Rendering Elements]

(Visualizing 3D Buildings, Topography, and Lighting)

[cite: 1]

Annotations & Graphics Analysis

1. 3D Mesh Geometry (Terrain & Buildings)

Rendering Element: The underlying polygonal structure that gives shape to mountains, valleys, and photogrammetry-generated city buildings [cite: 1].

How it creates 3D space: The mesh geometry is the foundation of the 3D space. Instead of a flat 2D map, height mapping and procedural generation allow the software to render physical elevation. As the camera moves, the perspective updates relative to this geometry, creating a true sense of volume and spatial depth [cite: 1].

2. Texture Mapping (Satellite Imagery)

Rendering Element: High-resolution 2D photographs (satellite and aerial imagery) wrapped around the 3D mesh [cite: 1].

How it creates 3D space: Without textures, the globe would simply be a collection of gray, shaded polygons. Texture mapping applies realistic surface colors and details to the geometry [cite: 1]. By utilizing varying levels of detail (LOD), the graphics engine seamlessly maps sharper textures as the user zooms closer, reinforcing the illusion of a tangible, realistic world.

3. Light Sources, Shadows, & Atmospheric Scattering

Rendering Element: The simulation of the sun as a directional light source, the resulting cast shadows on the terrain, and the blue atmospheric halo around the Earth [cite: 1].

How it creates 3D space: Lighting is critical for depth cues. The shadows cast by mountains or buildings give the user an immediate, intuitive understanding of scale and elevation [cite: 1]. The atmospheric scattering on the horizon provides a realistic horizon line, separating the 3D earth from the void of space.

4. Reflections and Conclusions

Based on the analysis of both the HCI and Computer Graphics elements of the selected application, the following reflections address the core relationship between visual rendering and user experience.

Q1: How does the Computer Graphics quality (e.g., realistic lighting or 3D view) help or hinder the user's ability to interact with the software? [cite: 1]

The high-quality Computer Graphics in Google Earth significantly **helps** the user interact with the software by providing realistic spatial context and depth perception [cite: 1]. Realistic lighting and accurate 3D meshes allow users to naturally gauge the height of mountains or the layout of a city, making navigation intuitive because it mimics real-world physics and optics. However, it can **hinder** interaction if the user's hardware is not powerful enough; slow rendering times or low-resolution texture popping can disorient the user and make precise interactions, like clicking on a specific building, difficult and frustrating [cite: 1].

Q2: What is one simple HCI change you would make to improve the user interface? [cite: 1]

One simple HCI change to improve the user interface would be to introduce a persistent, small **2D minimap overlay** (a "snap-to-top-down" view) in the corner of the screen. While navigating complex 3D environments, users often lose their sense of direction and geographic context. A minimalist 2D map showing their current footprint would reduce spatial disorientation, allowing them to understand exactly where they are looking without having to zoom all the way out [cite: 1].

End of Assignment